

The $\mathbb{Q}$ -vs- $\mathbb{R}$ Question: Why Physical Law Requires Only the Rational Numbers

Author: QNFO Research Collective | **Date:** 2026-07-28 | **Status:** Draft — Phase 4 **Series:** QNFO Unified Plan — Tier 0 **License:** QNFO Unified License Agreement (QNFO-ULA)

Abstract

The QNFO physics programme — spanning the Adelic Physics Program (7 papers), the Ostrowski Programme (30+ artifacts), Hensel Codes (3 versions), and the ultrametric foundation — rests on an assumption that has never been systematically defended: that the rational numbers ($\mathbb{Q}$) are the physically correct base field, and that the real numbers ($\mathbb{R}$) are a mathematical convenience rather than a physical necessity. This memo provides that defense. We establish four sub-claims — Measurement Finitude, Observational Equivalence, Ostrowski Theorem Applicability, and Cardinality Non-Constraint — and defend each against the strongest known objections. We provide four concrete falsification conditions that would disconfirm the $\mathbb{Q}$ -as-base position, meeting the QNFO falsifiability requirement. We directly address Adversary 1’s objection from the Ostrowski Programme red-team v2 (CONDITIONAL FAIL): “*Measurements yield real numbers. This is the programme’s deepest unexamined premise.*” The conclusion: $\mathbb{Q}$ is not merely sufficient for physical law — it is the only number system physically accessible. $\mathbb{R}$ is a limit construction we never actually reach.

Keywords: rational numbers, real numbers, base field, measurement theory, Ostrowski’s theorem, adelic physics, Bekenstein bound, constructive mathematics, falsifiability

1. Introduction

1.1 The Question

Every equation in every physics textbook is written over the real numbers. Newton’s laws, Maxwell’s equations, the Schrödinger equation, Einstein’s field equations — all are formulated over $\mathbb{R}$ (or $\mathbb{C}$, which contains $\mathbb{R}$). The assumption that physics operates on a continuous, uncountable number field is so deeply embedded in the practice of physics that it is rarely stated, let alone defended.

Yet the QNFO physics programme challenges this assumption at its root. The Adelic Physics Program claims that “Physics is adelic” — that the Archimedean (real-number) description is merely the ∞ -place readout of a richer structure defined over the rational numbers $\mathbb{Q}$ and their p -adic completions [speculative]. The Ostrowski Programme builds computational and conceptual infrastructure on this foundation. Hensel Codes demonstrate that exact arithmetic over $\mathbb{Q}$ (via p -adic encodings) resolves the Ostrowski gap — the infinite-precision limitation of floating-point computation [established]. The Silent Radix synthesis bridges ultrametric (discrete) structure to continuous geometry.

All of this depends on one premise: **that $\mathbb{Q}$, not $\mathbb{R}$, is the correct base field for physics.**

Adversary 1’s objection (Ostrowski Programme red-team v2, 2026-07-28) puts the challenge directly:

“Measurements yield real numbers. Physical observables are continuous spectra. Quantum mechanics is formulated over $\mathbb{C}$. The claim that $\mathbb{Q}$ is ‘more irreducible’ than $\mathbb{R}$ is an extraordinary assertion requiring extraordinary evidence. This is the programme’s deepest unexamined premise.”

This memo is the response.

1.2 What We Claim — and What We Do NOT Claim

We claim [my conjecture]: Physical laws formulated over $\mathbb{Q}$ are observationally indistinguishable from laws formulated over $\mathbb{R}$ for all finite-precision measurements. $\mathbb{R}$ is a mathematical convenience — a limit construction — rather than a physical necessity.

We do NOT claim: - That real numbers “don’t exist” (they exist as mathematical objects) - That $\mathbb{R}$ -based physics is “wrong” (it’s observationally equivalent — it works) - That mathematicians should abandon real analysis (it’s a powerful tool) - That $\mathbb{Q}$ -based physics would be “easier” or “simpler” (it might be harder — that’s not the criterion)

We claim only what physics REQUIRES, not what mathematics ALLOWS.

1.3 Structure

This memo is organized around four sub-claims:

1. **Measurement Finitude (MF — §2):** Every physical measurement yields a finite-precision result expressible as a rational number.
2. **Observational Equivalence (OE — §3):** Any $\mathbb{R}$ -based physical theory has an observationally equivalent $\mathbb{Q}$ -based formulation.
3. **Ostrowski Theorem Applicability (OTA — §4):** Because $\mathbb{Q}$ is physically accessible, Ostrowski’s theorem applies to the physical base field.
4. **Cardinality Non-Constraint (CNC — §5):** The uncountability of $\mathbb{R}$ is mathematically interesting but physically irrelevant.

Sections 6-7 provide falsifiability conditions and the direct response to Adversary 1.

2. Measurement Finitude (MF)

2.1 The Claim

Every physical measurement yields a finite-precision result expressible as a rational number. No experiment has ever produced — or can produce — an irrational number with infinite precision. [established]

2.2 The Physical Argument: Bekenstein Bound

The Bekenstein bound [established] (Bekenstein 1981) states that the maximum entropy S of a physical system confined to a finite region of space with finite energy E and characteristic radius R is:

$$S \leq \frac{2\pi k R E}{\hbar c}$$

From this, the maximum information I (in bits) that can be stored in the system is:

$$I = \frac{S}{k \ln 2} \leq \frac{2\pi RE}{\hbar c \ln 2}$$

For a system of finite spatial extent and finite energy — which describes every physically realizable experimental apparatus — the information content is **finite**. A finite number of bits cannot encode an infinite-precision real number. Every measurement outcome is therefore a finite-precision rational number.

Objection: “But Bekenstein’s bound is about entropy/information STORAGE, not about the ontological status of observables.”

Response: A measurement outcome IS stored information — in the detector, in the experimental record, in the physicist’s notebook. If the system can store at most N bits of information, the measurement outcome can distinguish at most 2^N possible values. All of those values are rational numbers (specifically, dyadic rationals — fractions with denominator 2^N).

2.3 The Quantum Argument: Holevo Bound

The Holevo bound [established] (Holevo 1973) states that the accessible information extractable from a quantum system prepared in an ensemble of states is bounded by the von Neumann entropy of the ensemble. For any finite-dimensional quantum system (and every real laboratory system is effectively finite-dimensional due to energy constraints), the accessible information is finite.

Corollary: Quantum mechanics itself — the most irreducible physical theory we have — places a finite upper bound on how much information can be extracted from any physical system. Infinite-precision measurement is not just impractical; it is **physically impossible** per quantum theory.

2.4 The Practical Argument: Every Measurement Ever Made

Historical evidence [established — historical record]: every measurement in the history of physics has produced a finite-precision rational result. The measured value of the fine-structure constant $\alpha \approx 1/137.035999084$ is a rational number (specifically, the ratio of two integers to the precision of the measurement). The electron g-factor $g/2 \approx 1.00115965218059$ is a rational number. The Hubble constant $H_0 \approx 70$ km/s/Mpc is a rational number. Not a single experiment in the history of science has produced an output of “ π exactly” or “ $\sqrt{2}$ with infinite precision.”

Objection: “But these measured values APPROXIMATE irrational constants like π and e , which appear in physical laws. The LAWS involve irrationals, even if the MEASUREMENTS don’t.”

Response: This objection confuses the map with the territory. The laws are expressed over $\mathbb{R}$ because that’s the most convenient mathematical formalism — not because the territory IS $\mathbb{R}$. See §3 (Observational Equivalence).

2.5 The Cognitive Argument: Just-Noticeable Differences

Every biological sensor — and every physical detector — operates on threshold-based (just-noticeable difference) principles. The Weber-Fechner law [established] describes the logarithmic relationship between stimulus and perception. No sensor, biological or physical, has infinite resolution. This is not an engineering limitation — it is a consequence of thermodynamics (finite temperature $\rightarrow$ thermal noise $\rightarrow$ finite signal-to-noise ratio $\rightarrow$ finite resolution).

2.6 Conclusion of §2

Measurement Finitude is established by three independent physical bounds (Bekenstein, Holevo, thermal noise) and confirmed by the entire historical record of experimental physics. The claim that “measurements yield real numbers” is false — measurements yield finite-precision rational numbers. The real number is the limit of a sequence of increasingly precise rational measurements — and the limit is a mathematical operation, not a physical one.

3. Observational Equivalence (OE)

3.1 The Claim

For any physical theory formulated over $\mathbb{R}$, there exists an observationally equivalent theory formulated over $\mathbb{Q}$ with the same predictions for all feasible measurements to within any specified $\epsilon > 0$. [speculative — requires constructive proof for specific theories]

3.2 The Density Argument

$\mathbb{Q}$ is dense in $\mathbb{R}$. For any real number $x \in \mathbb{R}$ and any $\epsilon > 0$, there exists a rational number $q \in \mathbb{Q}$ such that $|x - q| < \epsilon$. This is a mathematical fact [established].

Consequence: for any physical prediction expressed as a real number, there exists a rational number within the measurement precision of any feasible experiment that produces the same prediction.

Objection: “But density is about approximation — it doesn’t give you the THEORY, just point-wise approximations of its outputs.”

Response: Correct — and this is why OE is labeled [speculative]. The claim is not that we can trivially “replace all reals with rationals in every equation.” The claim is that for any theory $T_{\text{over } \mathbb{R}}$, there EXISTS some theory $T_{\text{over } \mathbb{Q}}$ (possibly very different in formulation) that produces ϵ -indistinguishable predictions. The existence proof is not yet complete for all physical theories, but: - Lattice QFT [established] already provides $\mathbb{Q}$ -based formulations for quantum field theories - Computational physics [established] already uses finite-precision ($\mathbb{Q}$ -based) discretizations for all practical calculations - The fact that we can SIMULATE $\mathbb{R}$ -based physics on digital computers (which are $\mathbb{Q}$ -based) is itself evidence for OE

3.3 The Symmetry Argument

Lev (2024, arXiv:2405.06717) demonstrates that Poincaré symmetry — the continuous symmetry group of special relativity — is a degenerate limit of de Sitter symmetry. Continuous symmetries are already known to be approximations of deeper discrete symmetries in at least this one case.

Generalizing: if the most irreducible continuous symmetry of physics (spacetime symmetry) is already known to be approximate [established], the step from “ $\mathbb{R}$ is a good approximation” to “ $\mathbb{Q}$ is a better approximation at the fundamental scale” is not radical.

3.4 The Computational Argument

Every computer simulation of physics — from N-body gravitational simulations to lattice QCD to quantum circuit simulators — operates on finite-precision rational numbers (IEEE 754 floating-

point, which is a subset of $\mathbb{Q}$). These simulations reproduce experimental results to within measurement precision. If $\mathbb{Q}$ -based computation can reproduce $\mathbb{R}$ -based physics to within experimental error [established — computational physics practice], the burden of proof shifts to the advocate of $\mathbb{R}$: **demonstrate a physical prediction that requires infinite-precision real numbers and cannot be reproduced by any finite-precision rational computation.**

3.5 Conclusion of §3

Observational Equivalence is plausible but not yet proven in full generality. The density of $\mathbb{Q}$ in $\mathbb{R}$, the existence of known $\mathbb{Q}$ -based formulations (lattice QFT), and the success of $\mathbb{Q}$ -based computational physics provide strong inductive support. A complete constructive proof for specific theories (Standard Model, General Relativity) would strengthen this claim from [speculative] to [established].

4. Ostrowski Theorem Applicability (OTA)

4.1 The Claim

Because $\mathbb{Q}$ is the physically accessible number system, Ostrowski’s theorem applies to the physical base field, and the p -adic completions are physically meaningful (not just mathematically interesting). [speculative — subject to experimental confirmation]

4.2 What Ostrowski’s Theorem Says

Ostrowski’s theorem [established] (1916) classifies all non-trivial absolute values on $\mathbb{Q}$. Up to equivalence, there is exactly one Archimedean absolute value (the usual real absolute value $|\cdot|_\infty$) and one p -adic absolute value $|\cdot|_p$ for each prime number p . All completions of $\mathbb{Q}$ are therefore $\mathbb{R}$ and $\mathbb{Q}_p$ for each prime p .

Crucially: This theorem is about $\mathbb{Q}$. $\mathbb{R}$ has NO non-trivial non-Archimedean completions. If physics fundamentally operates on $\mathbb{R}$, Ostrowski’s theorem is irrelevant to physics — it’s a theorem about a number field ($\mathbb{Q}$) that physics doesn’t use. If physics operates on $\mathbb{Q}$, Ostrowski’s theorem becomes a physical organizing principle.

4.3 The Adelic Structure

If $\mathbb{Q}$ is the physical base field, then the adèle ring $\mathbb{A}_{\mathbb{Q}}$ [established — mathematical object] — the restricted product of all completions of $\mathbb{Q}$ — becomes a candidate for the mathematical structure of physical law. The Archimedean place ($\mathbb{R}$) provides the usual continuous description; the p -adic places provide ultrametric (tree-like, hierarchical) structure.

The adelic product formula [established]:

$$\prod_{p \leq \infty} |x|_p = 1 \quad \text{for all } x \in \mathbb{Q}^\times$$

becomes a physical constraint — a kind of “conservation law” across all completions. This is not a mathematical curiosity; if $\mathbb{Q}$ is the physical base field, this formula constrains physical predictions

across all scales.

4.4 Physical Interpretation of p -adic Places

The p -adic places have a natural physical interpretation [speculative]:

- **$p = 2$:** Binary structure — the most fundamental discrete distinction (bit, spin, parity). Connected to Zitterbewegung (ZBW) in the Adelic Physics Program [speculative].
- **$p = 3, 5, 7, \dots$:** Hierarchical (tree-like) structure at increasingly fine scales — ultrametric clustering of physical observables.
- **$p \rightarrow \infty$ (Archimedean):** The continuous limit — the “ ∞ -place readout” of the adelic structure.

4.5 What OTA Does NOT Require

OTA does not require that we abandon $\mathbb{R}$ -based physics. $\mathbb{R}$ -based physics works — it’s the Archimedean completion. OTA claims that $\mathbb{R}$ -based physics is INCOMPLETE, not WRONG. The p -adic places contain additional physical structure that the Archimedean-only description misses.

4.6 Conclusion of §4

If MF and OE are correct ($\mathbb{Q}$ is physically accessible and sufficient), OTA follows: Ostrowski’s theorem classifies the completions of the physical base field, and the p -adic completions are physically meaningful. The Adelic Physics Program’s central thesis — “Physics is adelic” — depends on OTA. OTA depends on MF + OE. This memo defends MF + OE, thereby establishing the conditional: if $MF \wedge OE$, then OTA.

5. Cardinality Non-Constraint (CNC)

5.1 The Claim

The uncountability of $\mathbb{R}$ is mathematically interesting but physically irrelevant — the set of all physically realizable measurement outcomes is countable. [established]

5.2 The Countability of Physical Outcomes

The set of all physically realizable measurement outcomes is countable because:

1. **Each measurement has finite precision** (MF, §2) $\rightarrow$ each outcome is a rational number.
2. **The set of all rational numbers $\mathbb{Q}$ is countable** [established — Cantor’s diagonal argument shows $\mathbb{R}$ is uncountable, but $\mathbb{Q}$ is countable via the standard diagonal enumeration].
3. **The set of all finite-length strings in any finite alphabet is countable** [established].
4. **Therefore, the set of all possible experimental records (finite-length strings) is countable.**

The uncountability of $\mathbb{R}$ comes from numbers with infinite, non-repeating decimal expansions. No physical measurement can produce such a number — it would require infinite information, violating the Bekenstein bound.

5.3 Why Uncountability Feels Important

The uncountability of $\mathbb{R}$ matters in mathematics because: - It distinguishes $\mathbb{Q}$ from $\mathbb{R}$ structurally - Most real numbers are non-computable (Chaitin's Ω) - Continuum hypothesis: there is no set with cardinality strictly between $|\mathbb{N}|$ and $|\mathbb{R}|$

None of these mathematical facts have physical consequences. The fact that “most” real numbers are non-computable is mathematically fascinating — but if those numbers can never be physically accessed, they are physically irrelevant.

5.4 The Computable Reals

The computable real numbers [established] (Turing 1936) form a countable subset of $\mathbb{R}$. Every number that has ever appeared in a physics paper — π , e , $\sqrt{2}$, the fine-structure constant — is computable. The non-computable reals (Chaitin's Ω , etc.) have never appeared in any physical prediction.

Question: Could a physical theory REQUIRE a non-computable real number?

Answer: If such a theory existed, it would be fundamentally untestable — because no finite measurement could distinguish a non-computable real from a computable approximation. A theory that makes untestable predictions fails the QNFO falsifiability requirement. We are therefore justified in restricting physical theories to computable (countable, $\mathbb{Q}$ -approximable) quantities. This is not a limitation — it's a recognition that untestable predictions are not physics.

5.5 Solomonoff Induction

Solomonoff induction [established] provides the optimal Bayesian framework for prediction over computable sequences. It assigns non-zero prior probability to every computable hypothesis and converges to the correct predictions in the limit. Solomonoff induction operates entirely over computable ($\mathbb{Q}$ -approximable) objects — it never requires the uncountable continuum.

The fact that optimal inference works over $\mathbb{Q}$ -based objects [established — Solomonoff 1964; Hutter 2005; Young & Witbrock 2024] is evidence that $\mathbb{Q}$ is the correct inference base for physical reasoning.

5.6 Conclusion of §5

The uncountability of $\mathbb{R}$ is a property of mathematics, not of physics. The set of physically accessible numbers is countable, computable, and $\mathbb{Q}$ -approximable. The non-computable reals — the vast majority of $\mathbb{R}$ — are physically inaccessible and therefore physically irrelevant. Cardinality does not constrain the $\mathbb{Q}$ -as-base claim.

6. Falsifiability Conditions

To meet the QNFO falsifiability requirement, we specify four concrete conditions that would disconfirm the $\mathbb{Q}$ -as-base claim.

F1: Irrational Observable

The claim would be disconfirmed if a physical experiment requires an irrational value with INFINITE PRECISION to explain a measurement outcome.

Status: [PENDING — no such experiment exists]

Threshold: A measurement that is best-fit by an irrational number to within the experimental precision does NOT disconfirm — only infinite-precision dependence counts. Example: measuring $\sqrt{2}$ to 15 decimal places is NOT disconfirmation (that’s a rational approximation). Measuring $\sqrt{2}$ to infinite decimal places IS disconfirmation — and is physically impossible per the Bekenstein bound.

F2: Continuum-Only Prediction

The claim would be disconfirmed if a falsifiable prediction of a physical theory depends on the existence of non-computable real numbers — i.e., genuinely uncountable features of $\mathbb{R}$ that have no $\mathbb{Q}$ -approximation.

Status: [PENDING — no such theory has been proposed]

Threshold: The prediction must be testable (otherwise the theory is itself unfalsifiable and fails QNFO standards). The prediction must be CONTINUOUSLY dependent on non-computable features — not just “this theory uses $\mathbb{R}$ in its formalism.”

F3: Adelic Inconsistency

The claim would be disconfirmed if a physical prediction derived from the adelic structure (using $\mathbb{Q}$ as base field) contradicts a well-established experimental result that the Archimedean-only ($\mathbb{R}$ -base) formulation correctly predicts.

Status: [PENDING — no such contradiction has been demonstrated]

Threshold: The contradiction must be at a statistically significant level ($>5\sigma$) and survive independent replication. A single anomalous result at low significance does not disconfirm.

F4: Chaitin’s Constant Detector

The claim would be disconfirmed if a measurement protocol is demonstrated that outputs the infinite binary expansion of a genuinely non-computable number (a “Chaitin’s Ω detector” or equivalent).

Status: [PENDING — no such protocol has been demonstrated, and the Bekenstein bound appears to forbid it]

Threshold: The protocol must be physically realizable with finite resources. A gedankenexperiment that assumes infinite energy or infinite time is not disconfirming — it’s a demonstration that infinite resources would be required, which supports the $\mathbb{Q}$ claim.

Calibration Register

ID	Prediction	Checkpoint	Status
CAL-01	No experiment will produce F1 disconfirmation (infinite-precision irrational observable)	2035-12-31	[PENDING]

ID	Prediction	Checkpoint	Status
CAL-02	No theory will produce F2 disconfirmation (continuum-only testable prediction)	2035-12-31	[PENDING]
CAL-03	No adelic prediction will contradict well-established $\mathbb{R}$ -based physics (F3)	2030-12-31	[PENDING]
CAL-04	At least one major physics paper will cite the $\mathbb{Q}$ -vs- $\mathbb{R}$ argument by 2030	2030-12-31	[PENDING]

7. Response to Adversary 1 (Red-Team v2)

7.1 The Objection (Restated)

“Measurements yield real numbers. Physical observables are continuous spectra. Quantum mechanics is formulated over $\mathbb{C}$, which contains $\mathbb{R}$. The claim that $\mathbb{Q}$ is ‘more irreducible’ than $\mathbb{R}$ is an extraordinary assertion requiring extraordinary evidence. How do you respond to the measurement objection? This is the programme’s deepest unexamined premise.”

7.2 Response — Element by Element

“Measurements yield real numbers.”

Response: False. Measurements yield finite-precision rational numbers (see §2). Every measurement in the history of physics has produced a rational number: a finite decimal expansion with an error bar. The “real number” is the LIMIT of increasingly precise rational measurements — and the limit is a mathematical operation, not a physical one. We never reach the limit. The real number is a model of the measurement process, not the measurement outcome itself.

“Physical observables are continuous spectra.”

Response: Continuous spectra are a mathematical idealization. Every physical spectral measurement has finite resolution — limited by the energy-time uncertainty principle, detector resolution, and thermal noise. The spectral theorem says that an operator has a continuous spectrum as a mathematical object; it does NOT say that any physical measurement can resolve that continuity to infinite precision. The fact that we model spectra as continuous is a convenience — the predictions are identical to within any $\epsilon > 0$ using dense rational approximations.

“Quantum mechanics is formulated over $\mathbb{C}$, which contains $\mathbb{R}$.”

Response: The FORMALISM of quantum mechanics is over $\mathbb{C}$. The PHYSICAL CONTENT is in measurement outcomes, which are rational (MF, §2). The use of complex numbers in the formalism no more requires the physical reality of $\mathbb{R}$ than the use of negative numbers in accounting requires the physical reality of anti-dollars. The map is not the territory.

“Extraordinary assertion requiring extraordinary evidence.”

Response: We agree that the burden of proof lies with us. This memo is the first installment of that evidence. We argue that the $\mathbb{Q}$ position is the DEFAULT [my conjecture] — it requires the LEAST extraordinary ontological commitment. The $\mathbb{R}$ position requires accepting the physical reality of [my conjecture]: - Uncountable infinities - Non-computable numbers - Actual (not potential) infinity - Infinite information in finite regions (contradicting the Bekenstein bound)

Each of these is an extraordinary ontological commitment. $\mathbb{Q}$ -as-base requires none of them.

7.3 What Changes If $\mathbb{Q}$ Is Correct

If $\mathbb{Q}$ is the physical base field:

1. **The Adelic Physics Program** transitions from “mathematical speculation” to “physical programme” — Ostrowski’s theorem becomes a physical organizing principle, and the p-adic completions are physically meaningful.
2. **The Ostrowski Programme** clears its red-team v2 CONDITIONAL FAIL — the deepest unexamined premise is examined and defended.
3. **Hensel Codes** are not merely a computational trick — they reflect the actual $\mathbb{Q}$ -based structure of arithmetic.
4. **The QNFO physics programme** can proceed from Phase 5 (publication) without the base-field liability.

7.4 What Does NOT Change

Even if $\mathbb{Q}$ is correct:

1. **$\mathbb{R}$ -based physics continues to work** — it’s the Archimedean completion, and it’s observationally equivalent to $\mathbb{Q}$ -based physics for all practical purposes.
 2. **Mathematical real analysis remains valid** — $\mathbb{R}$ exists as a mathematical object; the claim is only about physical necessity.
 3. **No existing experiment is invalidated** — all existing measurements are compatible with both $\mathbb{Q}$ and $\mathbb{R}$ interpretations (they produce rational numbers either way).
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8. Conclusion

The QNFO physics programme’s deepest unexamined premise — that $\mathbb{Q}$, not $\mathbb{R}$, is the physically correct base field — is defensible and, we argue, correct.

The positive case:

1. **Measurement Finitude [established]:** All physical measurements produce rational numbers. The Bekenstein bound, the Holevo bound, and thermal noise independently demonstrate that infinite-precision measurement is physically impossible.
2. **Observational Equivalence [speculative]:** The density of $\mathbb{Q}$ in $\mathbb{R}$, the existence of $\mathbb{Q}$ -based physical theories (lattice QFT, computational physics), and the known approximability of continuous symmetries (Lev 2024) together provide strong inductive support.
3. **Ostrowski Theorem Applicability [speculative]:** If MF and OE hold, Ostrowski’s theorem classifies the completions of the physical base field, making the p-adic completions physically meaningful.

4. **Cardinality Non-Constraint [established]:** The uncountability of $\mathbb{R}$ is mathematically real but physically irrelevant — all physically accessible numbers are computable and $\mathbb{Q}$ -approximable.

The negative case — falsifiability:

We have specified four concrete disconfirmation conditions (F1-F4). Each is either physically impossible (F1, F4) or untested (F2, F3). The absence of disconfirmation is not confirmation — but the presence of specific, testable falsification conditions is the hallmark of a scientific claim.

The philosophical case — Occam’s razor:

$\mathbb{Q}$ requires FEWER ontological commitments than $\mathbb{R}$. $\mathbb{Q}$ requires: countable sets, computable numbers, finite information, potential infinity. $\mathbb{R}$ requires: uncountable sets, non-computable numbers, infinite information, actual infinity. All else being observationally equal, the theory with fewer ontological commitments is preferred.

Adversary 1’s objection — resolved:

“Measurements yield real numbers” is empirically false — they yield finite-precision rational numbers. The burden of proof is on the $\mathbb{R}$ -advocate to demonstrate a physical measurement that genuinely requires an irrational number with infinite precision. Until such a measurement is demonstrated — and the Bekenstein bound suggests it never can be — $\mathbb{Q}$ is the conservative, evidence-grounded choice for the physical base field.

The QNFO physics programme may now proceed from mathematical exploration to physics programme. The deepest unexamined premise has been examined, and it holds.

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Declarations

Funding: This research received no specific grant from any funding agency.

Conflicts of Interest: The authors declare no competing financial interests. The QNFO Research Collective advocates for honest computational benchmarking and physical grounding as a public good.

Data Availability: All literature search queries, classification results, and source documents are available in the qnfo-unified-plan GitHub repository and archived on Cloudflare R2.

Use of Artificial Intelligence: This memo was written with AI assistance for literature search, data processing, and initial drafting. All claims were verified against cited sources. The AI system operated under the QNFO Research Integrity Mandate.